

AC Steady State Analysis of Electric Circuits

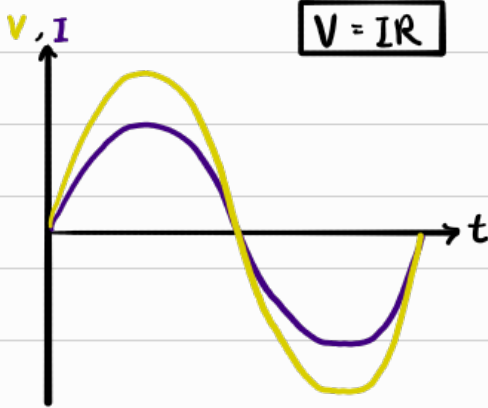
The analysis of RLC circuit transient & steady state response subject to sinusoid voltage & current sources.

Phasors & Complex Numbers

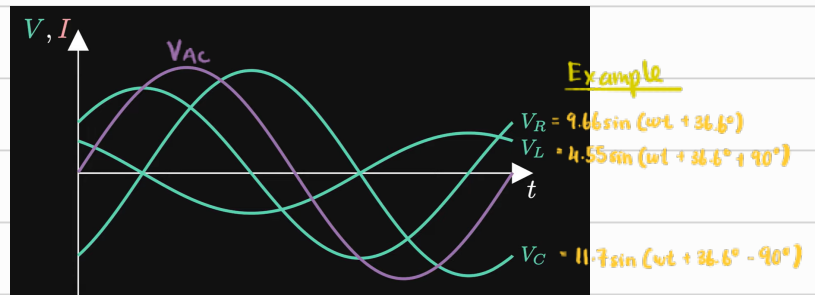
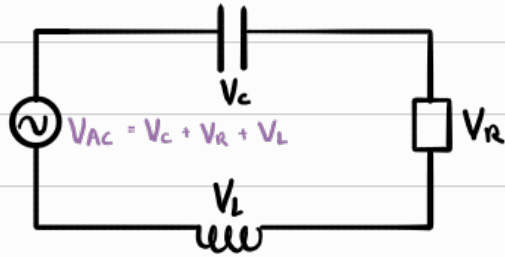
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In AC circuits, most relevant quantities (V, I) resemble sinusoidal waveforms.

For a simple circuit w only a resistor,

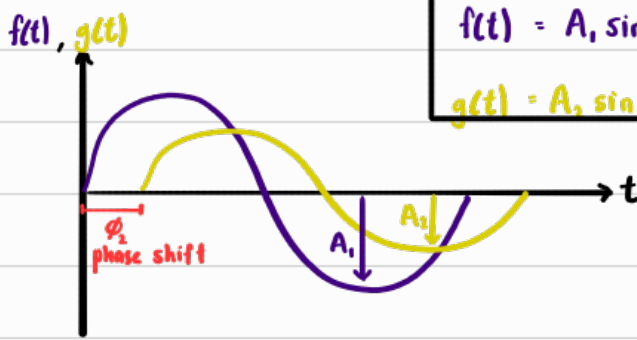


For circuits w ^{resist ΔV} capacitors & ^{resists ΔI} inductors



As u can see, if we were to find V_{AC} by summing up these trigo func., its gonna take forever. So we found a shortcut, phasors.

Phaser Explanation



$$f(t) = A_1 \sin(\omega t)$$

$$g(t) = A_2 \sin(\omega t + \phi_2)$$

For any steady state

AC electricity

$$f_{\text{output}} = f_{\text{input}}$$

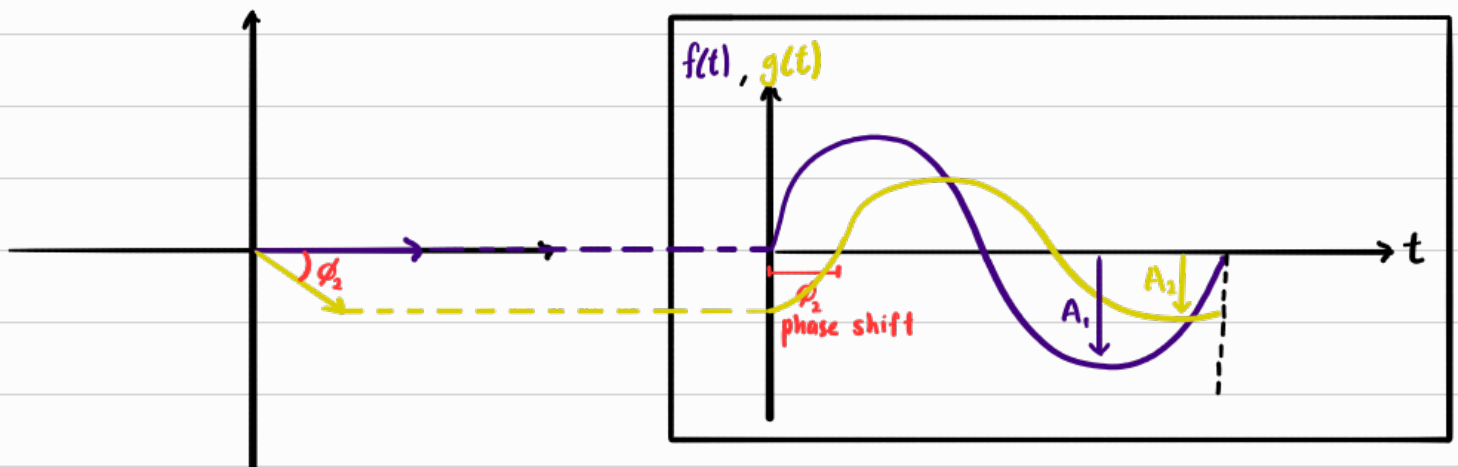
$$\Rightarrow \omega_{\text{output}} = \omega_{\text{input}}$$

Since $\omega_{\text{output}} = \omega_{\text{input}}$, our AC waves only have 2 parameters

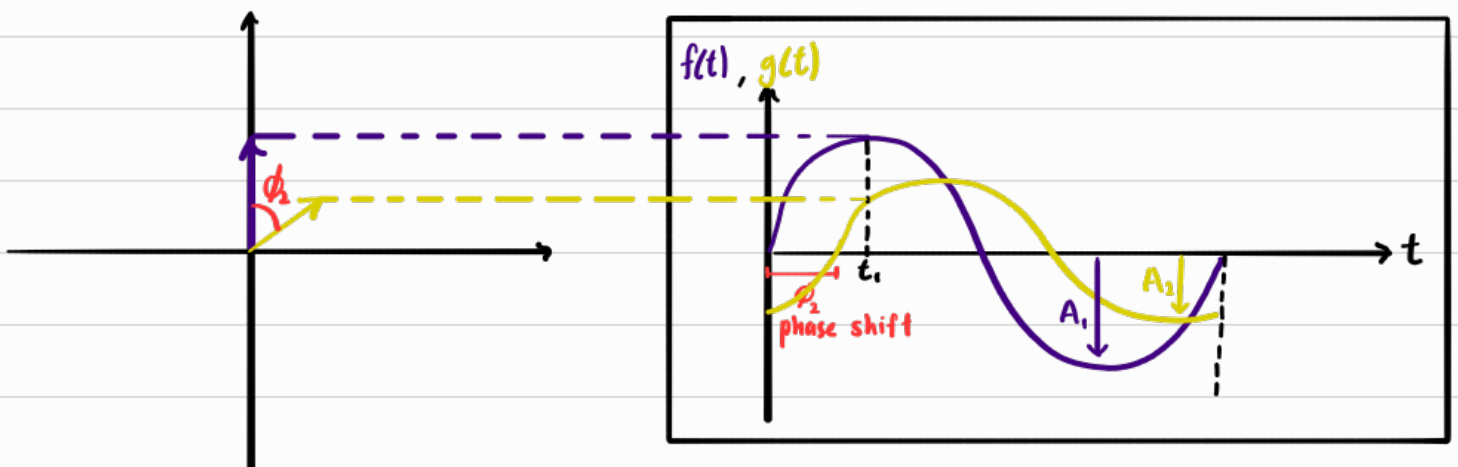
- 1) Amplitude, A
 - 2) Phase shift, ϕ
- } 2 Dimensions

$\therefore$ We can represent them w a 2D vector [aka. phase vectors]

phasors



If we rotate anticlockwise, the height of the vectors will plot out the wave [Vector Length doesn't change, since Vector length = Amplitude]

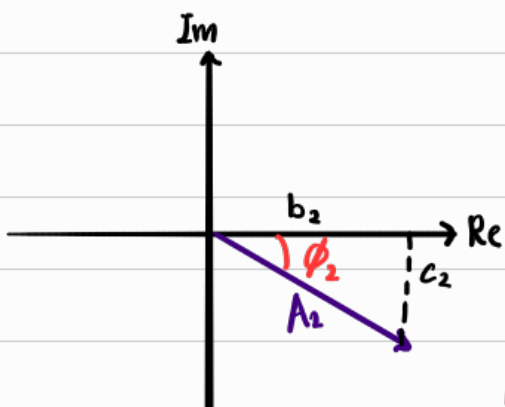


Do notice that since ω is constant [in S.S], the relative ϕ_2 bwn them never changes. So as t varies, $f(t)$ & $g(t)$ still represents the value of the wave at the same x-coordinate, t .

This means time is completely irrelevant. We can fully recreate our waveform as long as we know the amplitude, A & the phase shift, ϕ .

The 2D vectors we drew can be represented as complex numbers

Let $g(t) = b_2 + j c_2$



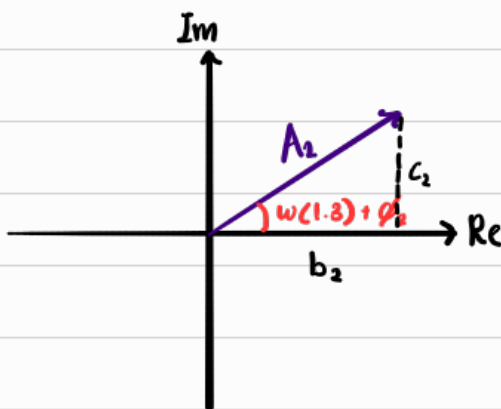
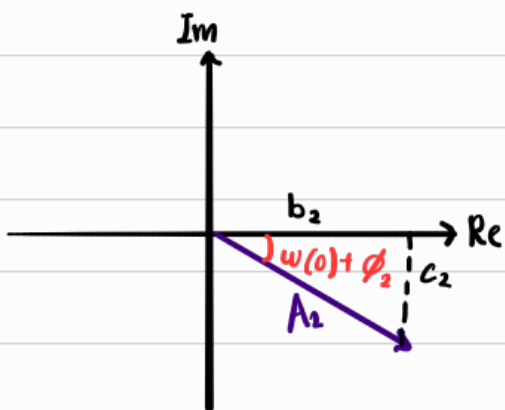
$$\begin{aligned} b_2 &= A_2 \cos(\phi_2) \\ c_2 &= A_2 \sin(\phi_2) \\ A_2 &= \sqrt{(b_2)^2 + (c_2)^2} \end{aligned}$$

$g(t) = c_2 = A_2 \sin(\phi_2)$
 [This is $g(0)$, $t = 0$]

We got this by knowing that its the height of the vector, c_2 that gives us the value at $t = 0$

But this just gives us $g(t)$ at $t = 0$ [$g(t) = A_2 \sin(\omega t + \phi_2) = A_2 \sin(\phi_2)$]

If we want to reintroduce t into our diagram, we just add ωt to the angle



$$\begin{aligned} b_2 &= A_2 \cos(\omega t + \phi_2) \\ c_2 &= A_2 \sin(\omega t + \phi_2) \end{aligned}$$

Now we get the complete waveform, $g(t) = A_2 \sin(\omega t + \phi_2)$. As t varies, it plots out the sinusoidal waveform.

3 ways expressing phasors

1) Complex numbers $[b + j c]$: $A \cos(\omega t + \phi) + j A \sin(\omega t + \phi)$

Addition & Subtraction

$$(b_1 + j c_1) \pm (b_2 + j c_2) = (b_1 \pm b_2) + j (c_1 \pm c_2)$$

Euler

ω is constant, t is an independent variable
 $\therefore$ This term is the same for all components in that circuit

2) Exponential

Diff & Int
 $\frac{d}{dt} A e^{j(\omega t + \phi)} = (j\omega) A e^{j(\omega t + \phi)}$
 $\int A e^{j(\omega t + \phi)} = \frac{1}{j\omega} A e^{j(\omega t + \phi)}$

$$: A e^{j(\omega t + \phi)} = [A e^{j\phi}] [e^{j\omega t}]$$

when $t = 0$

3) Polar coordinates $[A \angle \phi]$

Multiplication & Division

$$A_1 \angle \phi_1 \cdot A_2 \angle \phi_2 = A_1 A_2 \angle (\phi_1 + \phi_2)$$

$$A_1 \angle \phi_1 / A_2 \angle \phi_2 = A_1 A_2 \angle (\phi_1 - \phi_2)$$

$$: A \angle \phi$$

Convention

By convention, θ represent the counterclockwise angle made w the positive horizontal axis. Thus $\theta = 0$ represents an arrow lying perfectly flat to the right of the horizontal axis.

The horizontal projection [shadow] is maximum when the arrow is lying flat on the horizontal axis [$\theta = 0$]. Thus the trigo function that also gets its maximum value at $\theta = 0$ is cosine

$\therefore$ By convention cosine is the standard Phasor Reference

$$A \angle \phi = A \cos(\omega t + \phi)$$

Example :

What is $\sin(\omega t + \phi)$ in phasor ?

$$\begin{aligned}\sin(\omega t + \phi) &= \cos(\omega t + \phi - 90^\circ) \\ &= \angle \phi - 90^\circ\end{aligned}$$

$\therefore$ This means $\sin(\omega t + \phi)$ is 90° lagging/behind the reference (cosine)

Impedance

Previously, we learned that $V = IR$, where R is the opposition to current that results energy being lost as heat.

However, now in AC circuits, voltage changes causes current to change. There are components that act as ^{Capacitors & Inductors} temporary energy storage. Because it takes time to store and release, this creates a timing delay (phase shift) btwn the voltage's push & current flow.

Capacitors [I leads V by 90°]

$$I = C \frac{dV}{dt}$$

$$I(t) = (V_0 \omega C) \sin(\omega t + 90^\circ)$$

$$V(t) = V_0 \sin(\omega t)$$

- 1) I is just the rate of flow of charge, $\frac{dQ}{dt}$, but it still physically takes time for those charges to travel & actually accumulate on the plates
- 2) Since V is only created after those charges are present on the plates, this causes a delay [Phase diff, ϕ]
- 3) The charges has to actually travel to the plate 1st, and only after that period of time does the V change accordingly.

[Limiting factor: Charge travel time. That's why V only can change gradually]

V lead I by 90°

Inductors [I lag V by 90°]

$$V = L \frac{dI}{dt}$$

$$I(t) = \frac{V_0}{\omega L} \sin(\omega t - 90^\circ)$$

$$V(t) = V_0 \sin(\omega t)$$

- 1) The instant charges try to move [current] becu of a V , they create a magnetic field that fights back that flow [Lenz Law, $\mathcal{E} = - \frac{d\phi_B}{dt}$]
- 2) Since the current can only flow freely after the magnetic field stops fighting the initial flow, this causes a delay [Phase diff, ϕ]
- 3) Thus the V happens first, then only after waiting for the magnetic field to stop fighting back, the current can only change correspondingly

[Limiting factor: Opposing $\mathcal{E}$ caused by ΔV . That's why V can change instantaneously but the opposing $\mathcal{E}$ only allows I to change gradually.]

This opposition of current caused by the time it takes to build up E -field or B -fields, which forces V and I out of sync is called Reactance, X

Thus, the total opposition to current = $R + X$

$$\underline{Z} = R + X$$

$\therefore$ The actual complete ohm's law is $\vec{V} = I \underline{Z}$

Meaning of impedance $\underline{Z}$

$$\underline{Z} = \vec{V} / \vec{I} \text{ — phasors}$$

Impedance is the quantity that map the current phasor, $\vec{I}$ onto the voltage phasor, $\vec{V}$ telling us how much we have to stretch & rotate $\vec{I}$ to get $\vec{V}$.

$\therefore \underline{Z}$ is a polar quantity, $|Z| \angle \phi$ (OR) $b + jc$
 how much to stretch $\vec{I}$ to match the length of $\vec{V}$ how much to rotate $\vec{I}$ to match $\vec{V}$

Multiply j means $+90^\circ$
 Multiply $(-j)$ means -90°

Since I lead V by 90° , we have to rotate $\vec{I}$ CW by 90° to match $\vec{V}$

Since I lag V by 90° , we have to rotate $\vec{I}$ ACW by 90° to match $\vec{V}$

Resistor

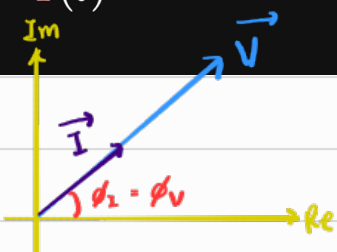
$$Z_R = R$$

$$V = V_0 e^{j\omega t}$$

$$I(t) = \frac{V(t)}{R}$$

$$I(t) = \frac{V_0}{R} e^{j\omega t}$$

$$\frac{V(t)}{I(t)} = R$$



Capacitor

$$Z_C = \frac{1}{\omega C} \angle -90^\circ$$

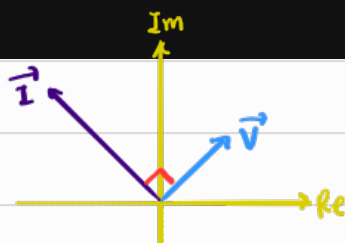
$$Z_C = -j \frac{1}{\omega C}$$

$$V = V_0 e^{j\omega t}$$

$$I(t) = C \frac{dV(t)}{dt}$$

$$I(t) = j\omega C V_0 e^{j\omega t}$$

$$\frac{V(t)}{I(t)} = -j \frac{1}{\omega C}$$



Inductor

$$Z_L = \omega L \angle 90^\circ$$

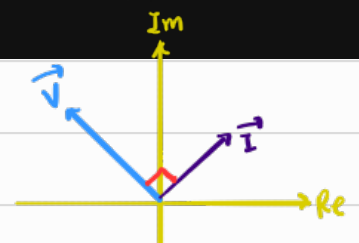
$$Z_L = j\omega L$$

$$V = V_0 e^{j\omega t}$$

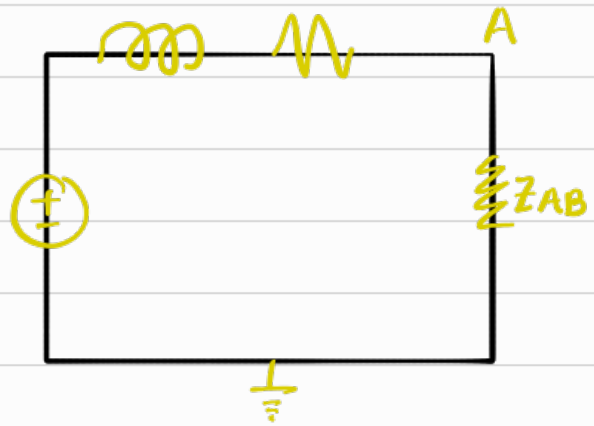
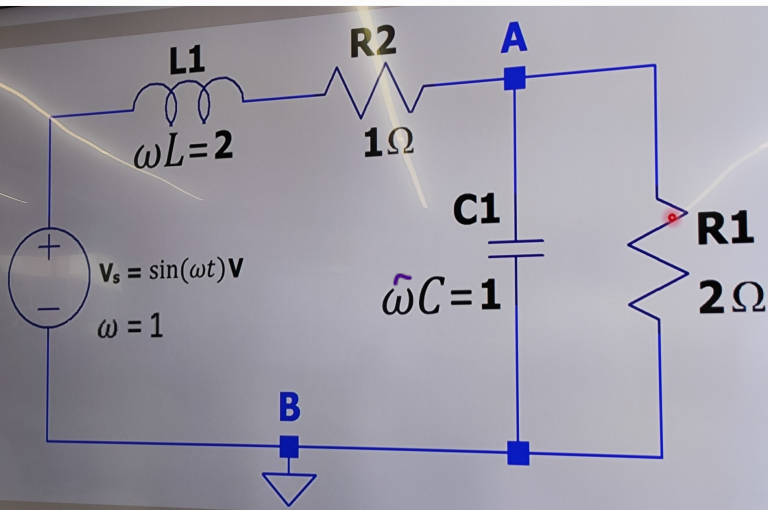
$$I(t) = \frac{1}{L} \int V(t) dt$$

$$I(t) = \frac{1}{j\omega L} V_0 e^{j\omega t}$$

$$\frac{V(t)}{I(t)} = j\omega L$$



Workshop Example 1



Find the following quantities: Z_{C1} , Z_{AB} , Z_{total} , V_{C1}

$$Z_{C1} = -j \left(\frac{1}{\omega C} \right) = -j \left(\frac{1}{1} \right) = -j \quad \#$$

$$\frac{1}{Z_{AB}} = \frac{1}{Z_{C1}} + \frac{1}{Z_{R1}}$$

$$\frac{1}{Z_{AB}} = \frac{Z_{R1} + Z_{C1}}{(Z_{C1})(Z_{R1})}$$

$$Z_{AB} = \frac{(Z_{C1})(Z_{R1})}{Z_{C1} + Z_{R1}}$$

$$= \frac{(-j)(2)}{(-j + 1)}$$

$$= 0.4 - 0.8j \quad [\text{use calc.}] \quad \#$$

$$Z_{total} = Z_{L1} + Z_{R2} + Z_{AB}$$

$$= (j\omega L) + 1 + (0.4 - 0.8j)$$

$$= j(2) + 1.4 - 0.8j$$

$$= 1.4 + 1.2j \quad \#$$

$$I_{AB} = I_{total}$$

$$\frac{V_{AB}}{Z_{AB}} = \frac{V_s}{Z_{total}}$$

$$V_{AB} = \frac{Z_{AB}}{Z_{total}} \times V_s$$

$$= \frac{(0.4 - 0.8j)}{(1.4 + 1.2j)} \times [1 \angle -90^\circ]$$

$$= -0.4706 + 0.1176j \quad V \quad [\text{use calc}]$$

$$V_{C1} = 0.485 \angle 165.97^\circ \quad [\text{use calc}]$$

$$V_s = \sin(\omega t)$$

$$= \cos(\omega t - 90^\circ)$$

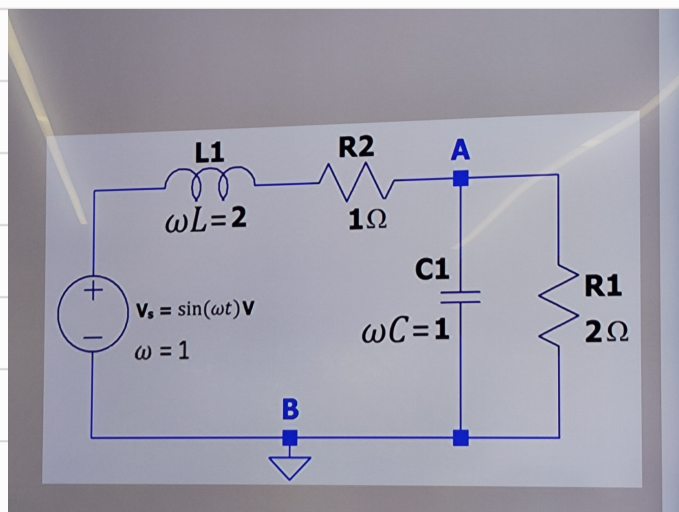
$$= 1 \angle -90^\circ$$

Note:

Z final ans in complex form

V, I final ans in phasor form

Workshop Example 1 continuation



Find the following quantities: I_{C1} , I_{R1} , I_{R2} , I_{L1}

$$\begin{aligned} I_{C1} &= V_{C1} / Z_{C1} \\ &= 0.485 \angle 166^\circ / (-j) \\ &= 0.485 \angle -104^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} I_{R2} &= I_{C1} + I_{R1} \\ &= 0.485 \angle -104^\circ + 0.243 \angle 166^\circ \\ &= 0.542 \angle -131^\circ \text{ A} \quad [\text{use calc}] \end{aligned}$$

$$\begin{aligned} I_{R1} &= V_{C1} / Z_{R1} \\ &= (0.485 \angle 166^\circ) / 2 \\ &= 0.243 \angle 166^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} I_{L1} &= I_{R2} \\ &= 0.542 \angle -131^\circ \text{ A} \end{aligned}$$

time domain

Find the following quantities: V_{R2} , V_{L1} , V_s , $V_{C1}(t)$

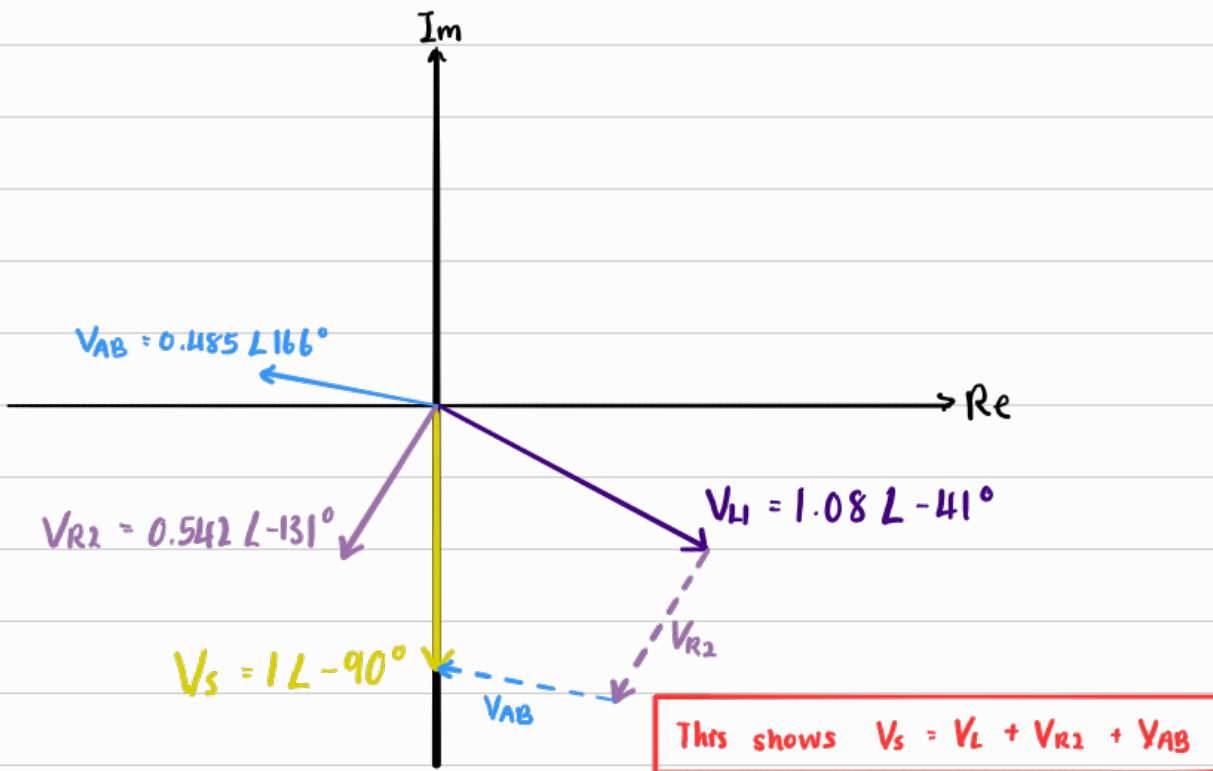
$$\begin{aligned} V_{R2} &= (I_{R2})(Z_{R2}) \\ &= (0.542 \angle -131^\circ)(1) \\ &= 0.542 \angle -131^\circ \text{ V} \end{aligned}$$

$$\begin{aligned} V_s &= V_L + V_{R2} + V_{AB} \\ &= (1.108 \angle -41^\circ) + (0.542 \angle -131^\circ) + (0.485 \angle 166^\circ) \\ &= 1 \angle -90^\circ \text{ V} \end{aligned}$$

$$\begin{aligned} V_{L1} &= (I_{L1})(Z_{L1}) \\ &= (0.542 \angle -131^\circ)(j\omega L) \\ &= (0.542 \angle -131^\circ)(2j) \\ &= 1.08 \angle -41^\circ \text{ V} \end{aligned}$$

$$\begin{aligned} V_{C1}(t) &= V_{AB} \\ &= 0.485 \angle 166^\circ \\ &= 0.485 \cos(\omega t + 166^\circ) \end{aligned}$$

Plotting V of workshop example 1



From pre-workshop vids

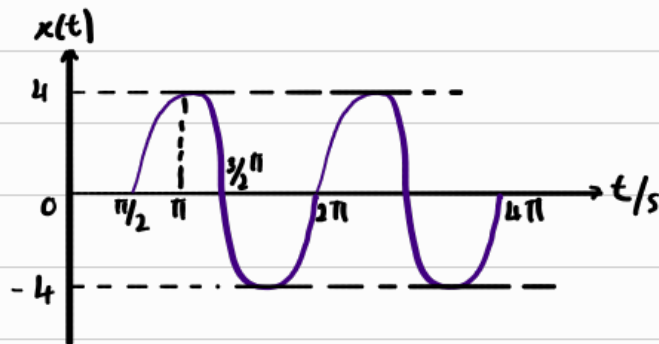
Sinusoids Revision

$$x(t) = A \sin(\omega t + \theta)$$

Amplitude

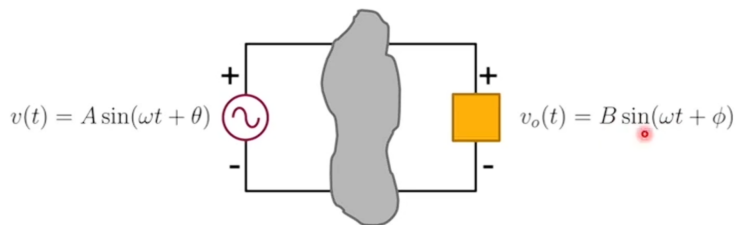
Angular Freq.
[$\omega = 2\pi f$]

$$x(t) = 4 \sin(2t - 90^\circ)$$



For a black box linear network

- For a 'black box' linear network:



- So, for a sinusoidal input we **know** the output will be another sinusoid, with the **same frequency**
 - The solution will involve determining B and ϕ

Since all we need to solve for is amplitude & phase, the phasor is perfect for it

Phasor Representation

$$x(t) = X_m \cos(\omega t + \theta)$$

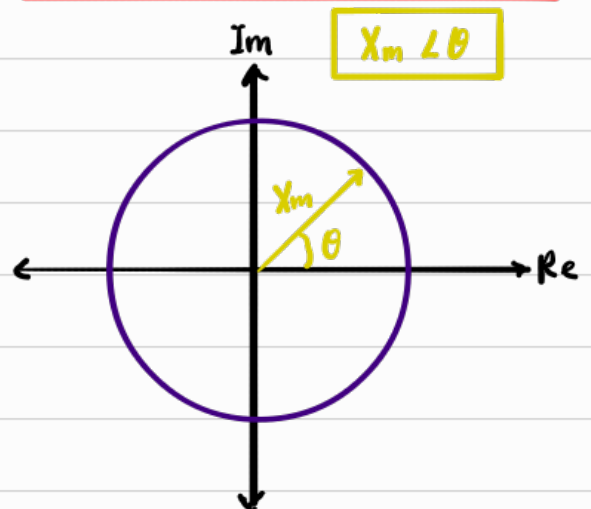
Recall Euler's formula, $e^{j\theta} = \cos\theta + j\sin\theta$

$$\therefore \cos(\omega t + \theta) = \text{Re}\{X_m e^{j(\omega t + \theta)}\}$$

$$= \text{Re}\{X_m e^{j\theta} (e^{j\omega t})\}$$

Constant amp. & phase part
[$X_m e^{j\theta}$]

Time varying part
[$e^{j\omega t}$]



Phasor for $x(t)$

$$\underline{X} = X_m \angle \theta = X e^{j\theta} = X \cos\theta + jX \sin\theta$$

Cartesian Form: $x + jy$

$$x = X_m \cos(\theta), \quad y = X_m \sin(\theta)$$

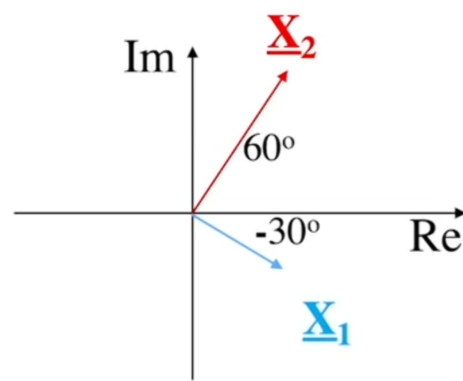
$$\theta = \tan^{-1}(y/x)$$

Example 6.2 (sinusoid sum)

$$x_1(t) = 20 \cos(\omega t - 30^\circ)$$

$$x_2(t) = 40 \cos(\omega t + 60^\circ)$$

$$\text{Find } x(t) = x_1(t) + x_2(t)$$



$$\underline{X} = \underline{X}_1 + \underline{X}_2$$

$$= 20 \angle -30^\circ + 40 \angle 60^\circ$$

$$= 20 e^{j(-30^\circ)} + 40 e^{j60^\circ}$$

$$= 20 [\cos(-30^\circ) + j \sin(-30^\circ)] + 40 [\cos(60^\circ) + j \sin(60^\circ)]$$

$$= 20 [\cos(30^\circ) - j \sin(30^\circ)] + 40 [\cos(60^\circ) + j \sin(60^\circ)]$$

$$= 37.3 + j(24.6)$$

$$= \sqrt{37.3^2 + 24.6^2} \angle \tan^{-1}\left(\frac{24.6}{37.3}\right)$$

$$x(t) = 44.7 \angle 33.4^\circ$$

$$= 44.7 \cos(\omega t + 33.4^\circ)$$

Example 6.3 (sinusoid decomposition/resolution)

$$x(t) = \sqrt{2} \cos(\omega t + 45^\circ)$$

$$\underline{X} = \sqrt{2} e^{j45^\circ}$$

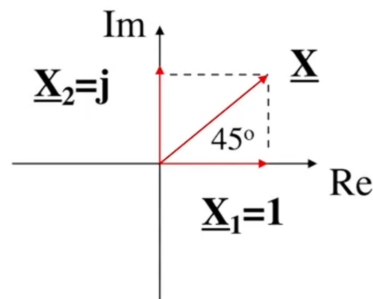
$$= \sqrt{2} \cos(45^\circ) + j \sqrt{2} \sin(45^\circ)$$

$$= \sqrt{2} \left(\frac{1}{\sqrt{2}}\right) + j \sqrt{2} \left(\frac{1}{\sqrt{2}}\right)$$

$$= 1 + j$$

$$= 1 \angle 0^\circ + 1 \angle 90^\circ$$

Vec. of length 1 pointing at angle 0°



$$x(t) = 1 \cos(\omega t + 0^\circ) + 1 \cos(\omega t + 90^\circ)$$

$$= \cos(\omega t) + \sin(\omega t)$$

In General

$$x(t) = X_m \cos(\omega t + \theta)$$

$$\underline{X} = a + jb = a \angle 0^\circ + b \angle 90^\circ$$

$$\text{Since } a = X_m \cos \theta, b = X_m \sin \theta \Rightarrow \theta = \tan^{-1}(b/a)$$

$$\Rightarrow x(t) = a \cos(\omega t) - b \sin(\omega t)$$

Phasor math basics:

Addition / subtraction

- Addition of two sinusoids

- Not straight forward in Polar ... $A\angle\theta + B\angle\phi = ?$

- Easy in Cartesian ...

$$(a + jb) + (c + jd) = (a + c) + j(b + d)$$

- So, convert polar to complex for +/- ...

$$\begin{aligned} A\angle\theta + B\angle\phi &= \\ &= A\cos(\theta) + jA\sin(\theta) + B\cos(\phi) + jB\sin(\phi) \\ &= [A\cos(\theta) + B\cos(\phi)] + j[A\sin(\theta) + B\sin(\phi)] \\ A\angle\theta - B\angle\phi &= \\ &= A\cos(\theta) + jA\sin(\theta) - B\cos(\phi) - jB\sin(\phi) \\ &= [A\cos(\theta) - B\cos(\phi)] + j[A\sin(\theta) - B\sin(\phi)] \end{aligned}$$

Phasor math basics:

Multiplication / division

- Multiplying two sinusoids

- Not super quick in Cartesian

- Easy in Polar

$$(A\angle\theta) \times (B\angle\phi) = AB\angle(\theta + \phi)$$

- Dividing two sinusoids

- Not straightforward in Cartesian

- Easy in polar $\frac{A\angle\theta}{B\angle\phi} = \frac{A}{B}\angle(\theta - \phi)$

Phasor Domain Ohm's Law

$$\underline{V} = \underline{Z} \underline{I}$$

complex impedance

Time domain

$$v(t) = V_m \cos(\omega t + \theta_v)$$

$\longleftrightarrow$

Phasor Domain

$$\underline{V} = V_m \angle \theta_v$$

$$i(t) = I_m \cos(\omega t + \theta_i)$$

$\longleftrightarrow$

$$\underline{I} = I_m \angle \theta_i$$

$$\text{Since } \underline{V} = \underline{Z} \underline{I}$$

$$\underline{Z} = \frac{V_m}{I_m} \angle (\theta_v - \theta_i) \, \Omega$$

Complex Impedance $\underline{Z}$ of components

Resistor : $\underline{Z} = R$

Inductor : $\underline{Z} = j\omega L$

Capacitor : $\underline{Z} = 1/j\omega C$
 $= -j/\omega C$

Note :

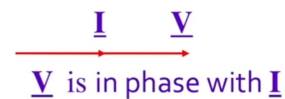
Multiplying j means $+90^\circ$ [lag]

" $-j$ means -90° [lead]

Comparison of $\underline{V}$ - $\underline{I}$ phasor relations for R, L, C (with $\underline{I}$ as reference)

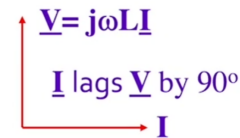
Resistor

$$\underline{V} = \underline{Z} \underline{I} = R \underline{I}$$



Inductor

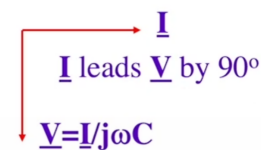
$$\underline{V} = \underline{Z} \underline{I} = j\omega L \underline{I}$$



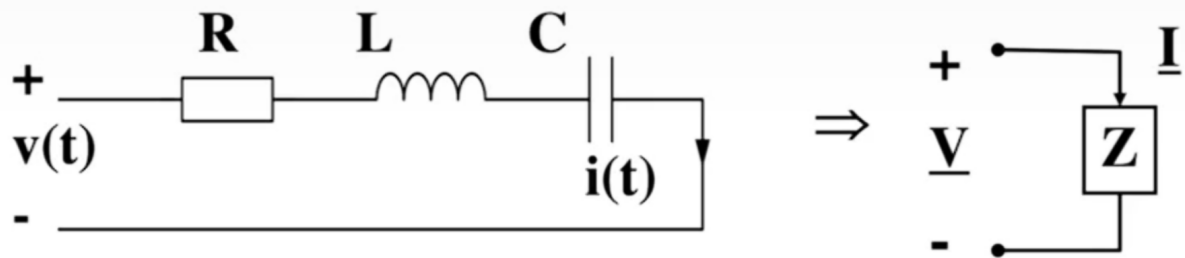
Capacitor

$$\underline{V} = \underline{Z} \underline{I} = \underline{I} / j\omega C$$

$$= (-j/\omega C) \underline{I}$$



Impedance of RLC combination/network



$$\underline{V} = \underline{V}_R + \underline{V}_L + \underline{V}_C = \underline{Z}\underline{I}$$

$$\underline{Z} = R + j\omega L + 1/j\omega C$$

$$= R + j(\omega L - 1/\omega C)$$

$$= R + jX(\omega) \Omega$$

• **Resistance:** R

• **Reactance:** $X(\omega) = \omega L - 1/\omega C$

• R and X can both be function of ω (Example 6.4)

• Phasor domain Ohm's law: $\underline{V} = \underline{Z}\underline{I} = (R + jX) \underline{I}$

Time $\Leftrightarrow$ Phasor

$$v(t) \Leftrightarrow \underline{V}$$

$$i(t) \Leftrightarrow \underline{I}$$

$$R \Leftrightarrow R$$

$$L di/dt \Leftrightarrow j\omega L \underline{I}$$

$$C dv/dt \Leftrightarrow j\omega C \underline{V}$$

Admittance, series/parallel combination

• **Admittance Y** is analogous to conductance in DC circuits:

How easily current flows

$$\underline{Y} = \frac{1}{\underline{Z}} = \frac{\underline{I}}{\underline{V}} \text{ (Siemens)}$$

How hard current flows

• Series/Parallel combinations are the same as for resistors

$$\underline{Z}_S = \underline{Z}_1 + \underline{Z}_2 + \cdots + \underline{Z}_n$$

$$\underline{Z}_P = \underline{Z}_1 \parallel \underline{Z}_2 \parallel \cdots \parallel \underline{Z}_n = \frac{1}{1/\underline{Z}_1 + 1/\underline{Z}_2 + \cdots + 1/\underline{Z}_n}$$

• Voltage and current division

The same as in DC case except that R_1 and R_2 are replaced by Z_1 and Z_2